Week #10 : Membrane Transport — Water Potential, Active Transport & Bulk Transport

Subject/الموضوع: AP Biology Unit Title: Cells / Membrane Transport Grade12: الصف

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents المصادر و الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1	I can explain water potential using solute and pressure potential. I can calculate Ψ = ΨP + ΨS and predict net water movement from higher to lower water potential.	1.1 Flexible display of information (size, color, font, layout).	Week 10 PPT — Lesson 1: Water Potential; osmosis review, T-chart of solute/pressure potential, water-potential calculations and exit ticket. Review Teams/AP Biology/General channel
Lesson 2 الحصة الثانية	I can explain active transport as ATP-dependent movement against a concentration gradient. I can model the sodium–potassium pump and explain how it maintains an electrochemical gradient.	5.3 Build fluency with graduated scaffolds and models.	Week 10 PPT — Lesson 2: Active Transport; sodium–potassium pump comic-strip model, 3 Na^+ out / 2 K^+ in, membrane potential and AP-style question. Review Teams/AP Biology/General channel
Lesson 3 الحصة الثالثة	I can explain cotransport as coupled movement driven by an existing solute gradient. I can compare endocytosis and exocytosis, including phagocytosis, pinocytosis and receptor-mediated endocytosis.	6.3 Facilitate organization with graphic organizers, prompts, and note-taking guides.	Week 10 PPT — Lesson 3: Cotransport and Bulk Transport; cellular-detective case files, vesicle transport, AP-style question and exit ticket. Review Teams/AP Biology/General channel
Lesson 4 الحصة الرابعة	I can organize membrane transport mechanisms into a concept map showing their relationships. I can select and justify an appropriate transport mechanism for a given cellular scenario.	2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	Week 10 PPT — Lesson 4: Cell Membrane Transport Concept Map; term sorting, group poster, gallery walk, peer feedback and exit ticket. Review Teams/AP Biology/General channel
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	• Bring Biology notebook, calculator and poster paper for calculations, models and concept maps. • Review PPT on Teams. Reinforce direction of gradients, ATP use, vesicle transport and evidence-based explanations.		

الأسبوع# Week #10

Technology Integration دمج التكنولوجيا	• PowerPoint diagrams, calculations, sodium–potassium pump models and transport animations. • Microsoft Teams for study materials and review; poster concept mapping.
Cross-curricular links الروابط بين المواد الدراسية	• Mathematics: addition of positive/negative water-potential values and comparison of gradients. • Chemistry: ion concentrations, electrical charge, ATP and free energy. • UAE engineering: irrigation, pumps and transport networks.
External Assessment Integration دمج التقييمات الخارجية	AP Biology Unit 2: Cells — membrane transport; AP-style data interpretation, calculations, mechanistic explanations and model-based reasoning. The uploaded slides label IB subtopic 1.4 Membrane Transport.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	• Thinking: quantitative reasoning, classification, systems modeling and prediction. • Communication: annotated diagrams, comic strips, concept maps and peer feedback. • Collaboration: partner T-chart, group posters and gallery walk.
Standards المعايير	• Lesson 1: Water potential (Ψ = ΨP + ΨS), pressure/solute potential and net water movement. • Lesson 2: Active transport, sodium–potassium pump and electrochemical gradients. • Lesson 3: Cotransport and vesicle-mediated bulk transport. • Lesson 4: Synthesis of passive, active and bulk membrane transport. • Uploaded PPT subtopic: 1.4 Membrane Transport (lesson 4 also integrates 1.3 membrane structure).
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Lesson 1: Why does water move into some plant cells and out of others? Lesson 2: How can a cell pump substances against their concentration gradients? Lesson 3: How do cells transport large cargo or couple one solute's movement to another? Lesson 4: How do different membrane transport mechanisms work together?
Success Criteria معايير النجاح **(Measurable)**	Lesson 1 — Water Potential MUST: Define water, solute and pressure potential. SHOULD: Calculate Ψ and predict net water movement. COULD: Interpret experimental water-potential data. Lesson 2 — Active Transport MUST: Identify gradient direction, ATP use and carrier proteins. SHOULD: Sequence the sodium–potassium pump (3 Na^+ out, 2 K^+ in). COULD: Predict consequences of pump failure for membrane potential. Lesson 3 — Cotransport and Bulk Transport MUST: Define cotransport, endocytosis and exocytosis. SHOULD: Explain indirect energy use in cotransport and distinguish vesicle mechanisms. COULD: Predict the effects of bulk-transport failure.

	Lesson 4 — Transport Concept Map MUST: Sort membrane transport terms correctly. SHOULD: Connect mechanisms using accurate linking words. COULD: Justify the most suitable mechanism in a novel scenario.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Respect الاحترام Heritage التراث
	المجال Area: Innovation / Sustainability / UAE Applications
	• UAE drip irrigation and arid agriculture: connect soil/plant water potential to water movement. • Dubai fountains and high-rise water pumps: analogy for energy-dependent movement against a gradient. • Dubai public transport networks: analogy for coordinated cotransport and bulk transport. • Masdar smart-city planning: analogy for organizing transport mechanisms into a connected system.